

```
#include <stdio.h>
```

```
#include <time.h>
```

```
void constantTime(int n){
```

```
    int x=n*n;
```

```
}
```

```
void linearTime(int n){
```

```
    int sum=0;
```

```
    for(int i=0;i<n;i++){
```

```
        sum+=i;
```

```
    }
```

```
}
```

```
void quadraticTime(int n){
```

```
    int count=0;
```

```
    for(int i=0;i<n;i++){
```

```
        for(int j=0;j<n;j++){
```

```
            count++;
```

```
        }
```

```
    }
```

```
}
```

```
int main(){
```

```
    clock_t start,end;
```

```
    double time_taken;
```

```
    int sizes[]={1000,5000,10000};
```

```
    int len=sizeof(sizes)/sizeof(sizes[0]);
```

```
    for(int i=0;i<len;i++){
```

```
        int n=sizes[i];
```

```
        start=clock();
```

```
    constantTime(n);
    end=clock();
    time_taken=(double)(end-start)/CLOCKS_PER_SEC;
    printf("%d %f ",n,time_taken);
    start=clock();
    linearTime(n);
    end=clock();
    time_taken=(double)(end-start)/CLOCKS_PER_SEC;
    printf("%f ",time_taken);
    start=clock();
    quadraticTime(n);
    end=clock();
    time_taken=(double)(end-start)/CLOCKS_PER_SEC;
    printf("%f\n",time_taken);
}
return 0;
}
```